

AI WORKFORCE TRANSFORMATION ANALYTICS

Enterprise Decision Framework for AI Adoption Strategy

Quantifying automation potential, ROI, and phased deployment
roadmap across a simulated 2,932-person technology workforce

August 2024 | Decision Analytics Portfolio Project

All data is synthetic. Figures are illustrative only.

2,932	\$373M	29	145	\$58.5M	7.5 mo	\$108.7M	0.969
Total FTE	Annual Labour Cost	Roles Analysed	Tasks Decomposed	Est. Net Annual Savings	Payback (Moderate)	3-Year NPV (@10%)	Min Spearman p

01 | Executive Summary

Business Context

Management of a large technology company commissioned this analysis to answer three strategic questions: (1) Which roles and tasks have the highest potential for AI automation or augmentation? (2) Where should AI be deployed first to maximise ROI? (3) What phased implementation strategy reduces operating cost while controlling quality and operational risk?

Using a proprietary **AI Adoption Priority Score (AAPS)** framework, scenario financial modelling, and multi-variable sensitivity analysis, this report delivers a data-driven deployment roadmap with full financial justification.

Key Findings

#	Finding	Implication
1	Engineering (\$115.6M), Sales (\$61.4M), and Customer Success (\$47.9M) represent 61% of total labour cost.	AI transformation must target these three domains first to move the needle.
2	Automation Potential inversely correlates with salary ($r = -0.846$) and Decision Intensity ($r = -0.975$).	High-pay, high-judgment roles (Legal, PM, HR BP) should not be automated.
3	Only 5 roles qualify for Full Automation; 18 for AI Augmentation; 6 are Low AI Suitability.	Augmentation strategy (productivity uplift) outweighs headcount reduction in value.
4	Moderate scenario (65% adoption) yields \$58.5M net annual savings with a 7.5-month payback.	Recommended: best risk-adjusted efficiency versus Conservative or Aggressive.
5	AAPS rankings are robust: minimum Spearman $\rho = 0.969$ across 12 weight perturbations.	Top-5 Phase 1 priorities hold regardless of reasonable weight disagreements.

02 | Workforce Economics Baseline

Before investing in AI, we must understand where capital is currently deployed. The chart below shows total annual labour cost (including 30% benefits load) by department. Any AI strategy that does not target Engineering, Sales, or Customer Success has limited enterprise-level impact.

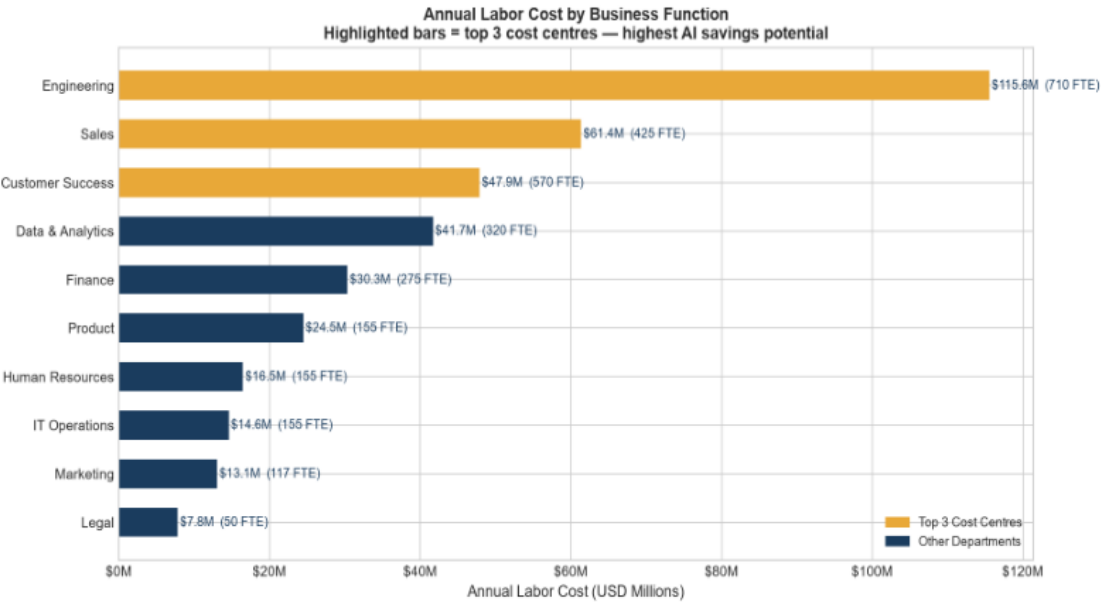


Fig 1: Annual Labour Cost by Department (\$M) — Engineering accounts for 31% of total spend

Department Cost Summary

Department	Roles	Headcount	Total_Cost_MUSD	Cost_Share_%
Engineering	3	710	115.57	31.0
Sales	3	425	61.36	16.4
Customer Success	3	570	47.89	12.8
Data & Analytics	3	320	41.69	11.2
Finance	4	275	30.35	8.1
Product	3	155	24.49	6.6
Human Resources	3	155	16.48	4.4
IT Operations	2	155	14.57	3.9
Marketing	3	117	13.06	3.5
Legal	2	50	7.83	2.1

03 | AI Automation & Augmentation Opportunity

We classify every task into: **Full Automation** (AI replaces human effort) or **AI Augmentation** (AI accelerates human output). Across the enterprise, we identified a **\$150.6M automatable cost pool** (40.3% of total) and a **\$180.4M augmentable cost pool** (48.3% of total).

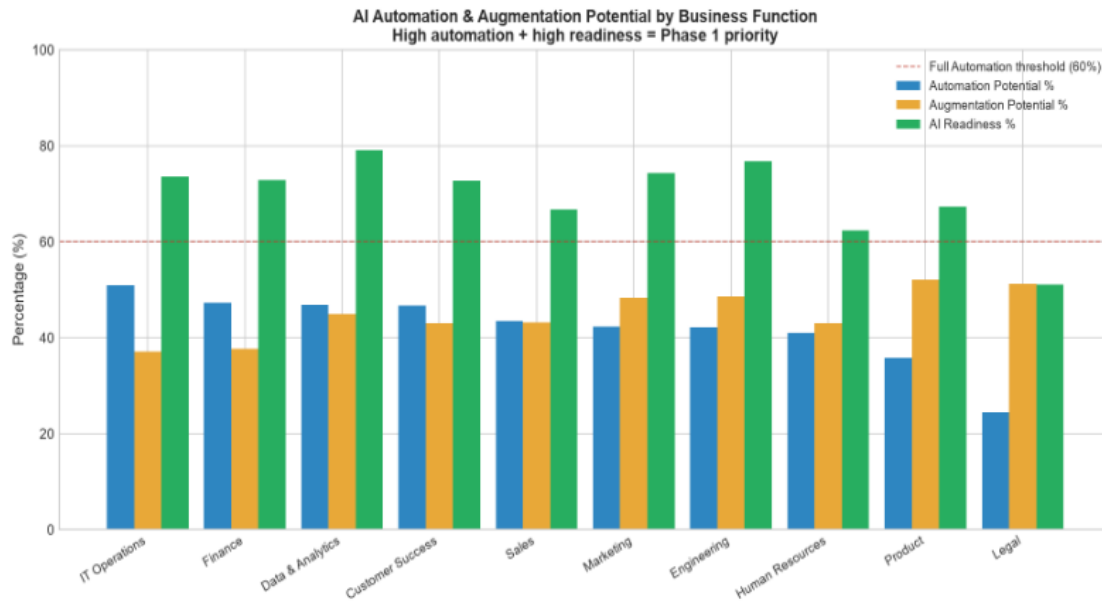


Fig 2: Avg Automation & Augmentation Potential by Department (%) — Finance and IT Operations lead in automation readiness

Productivity Improvement Under AI Augmentation

AI does not only cut costs — it dramatically improves output per FTE. The chart below shows estimated productivity uplift for each role under the moderate adoption scenario (65% capture rate).

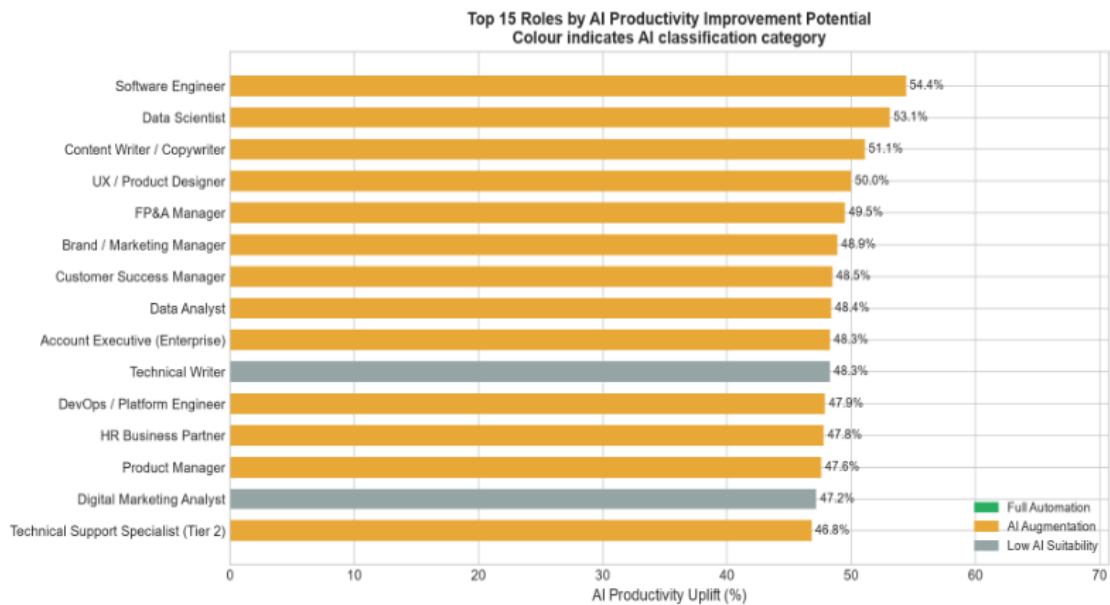


Fig 3: Estimated AI Productivity Uplift by Role (%) — Technical roles show transformational gains

Key Insight: Software Engineers show the highest absolute dollar impact from AI augmentation. While their automation potential is only moderate (due to high skill complexity), Copilot-style tools demonstrated 30–55% productivity uplift in published MIT/GitHub studies. At 420 FTE × \$130K salary, a 40% output gain is worth ~\$28M in annual productive capacity.

04 | AI Adoption Priority Score (AAPS) Framework

Scoring Methodology

To eliminate subjective decision-making, we developed the AAPS: a transparent, weighted composite score normalising six factors to the [0,1] range using min-max normalisation.

Factor	Weight	Rationale
Automation Potential	25%	Core feasibility — without this, ROI is impossible
Cost Saving Potential	25%	Direct value driver — scales automation savings to FTE count
Productivity Improvement	15%	Augmentation value — output uplift even where automation is low
Implementation Feasibility	15%	Risk-adjusted deployment likelihood within 24 months
Quality Risk (penalty)	10%	Downside modifier — high quality risk reduces AAPS
Human Judgment Req. (penalty)	10%	Downside modifier — roles needing judgment scored lower

Full AAPS Priority Ranking (All 29 Roles)

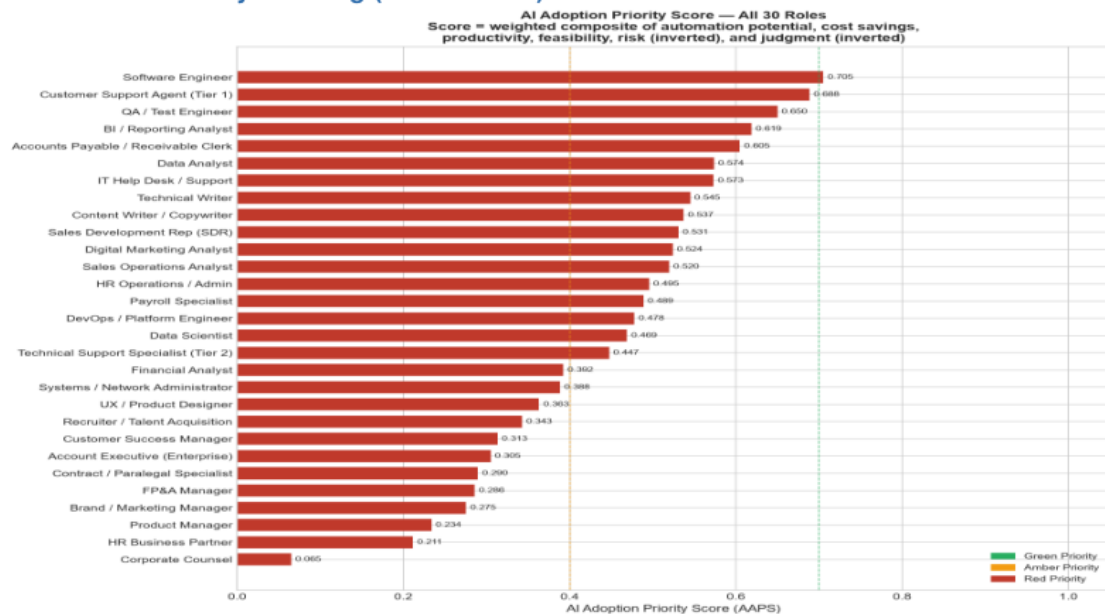


Fig 4: AAPS Priority Ranking — All 29 Roles (Green = Phase 1 Quick Wins, Red = Defer)

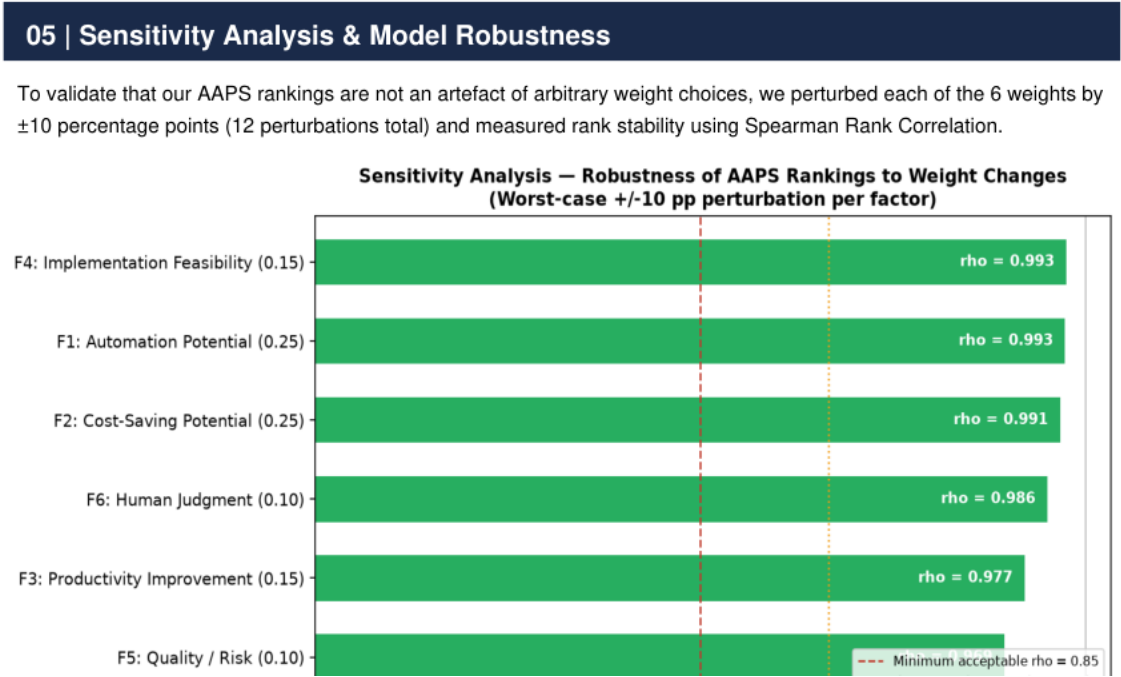
Top 10 Priority Roles by AAPS

Role	Department	AAPS	AI Classification	Phase
Software Engineer	Engineering	0.7051	AI Augmentation	Phase 1
Customer Support Agent (Tier 1)	Customer Success	0.6884	Full Automation	Phase 1
QA / Test Engineer	Engineering	0.6498	Low AI Suitability	Phase 1
BI / Reporting Analyst	Data & Analytics	0.6188	Full Automation	Phase 1
Accounts Payable / Receivable Clerk	Finance	0.6047	Full Automation	Phase 1
Data Analyst	Data & Analytics	0.5736	AI Augmentation	Phase 1

Role	Department	AAPS	AI Classification	Phase
IT Help Desk / Support	IT Operations	0.5731	Full Automation	Phase 1
Technical Writer	Product	0.5449	Low AI Suitability	Phase 1
Content Writer / Copywriter	Marketing	0.5371	AI Augmentation	Phase 1
Sales Development Rep (SDR)	Sales	0.5308	Low AI Suitability	Phase 1

05 | Sensitivity Analysis & Model Robustness

To validate that our AAPS rankings are not an artefact of arbitrary weight choices, we perturbed each of the 6 weights by ±10 percentage points (12 perturbations total) and measured rank stability using Spearman Rank Correlation.



Robustness Verdict

The minimum Spearman ρ of **0.969** (average: 0.988) confirms that the strategic conclusions are statistically robust. The top-5 Phase 1 roles remain in the top tier under every weight perturbation tested.

Role Stability Summary

Category	Roles	Max Rank Movement	Implication
Stable (top 5)	SW Engineer; CS Agent T1; QA Engineer; BI Analyst; AP/AR Clerk	0–3 positions	Phase 1 deployment is unconditional.
Sensitive (mid)	SDR; Content Writer; HR Ops Admin; Data Scientist; Payroll Specialist	4–5 positions	Phase 2/3 — monitor weight assumptions.
Stable (bottom)	Corporate Counsel; HR BP; Product Manager	0–1 positions	Confirmed: do NOT automate these roles.

06 | Scenario Analysis & Financial Modelling

We evaluated three adoption scenarios to bound the financial opportunity and guide the investment decision. The recommended **Moderate Scenario** (65% capture) balances speed of execution with operational risk management.

Scenario	Adoption Rate	Gross Labour Savings	Impl. Cost	Net Annual Benefit	Payback (Months)	3-Year NPV
Conservative	40%	\$24.1M	\$44.2M	\$36.0M	~14.7	\$54.7M
Moderate ★	65%	\$39.2M	\$36.9M	\$58.5M	~7.5	\$108.7M
Aggressive	85%	\$51.2M	\$33.2M	\$76.1M	~5.2	\$156.3M

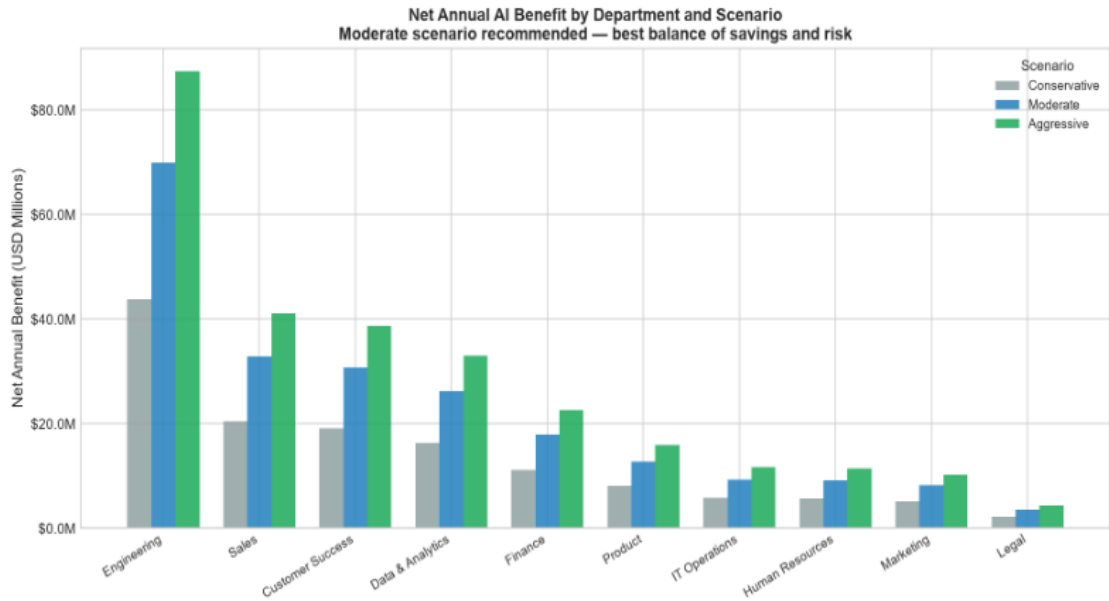


Fig 6: Scenario Savings by Department — Conservative vs Moderate vs Aggressive (\$M)

Savings vs Implementation Cost Analysis

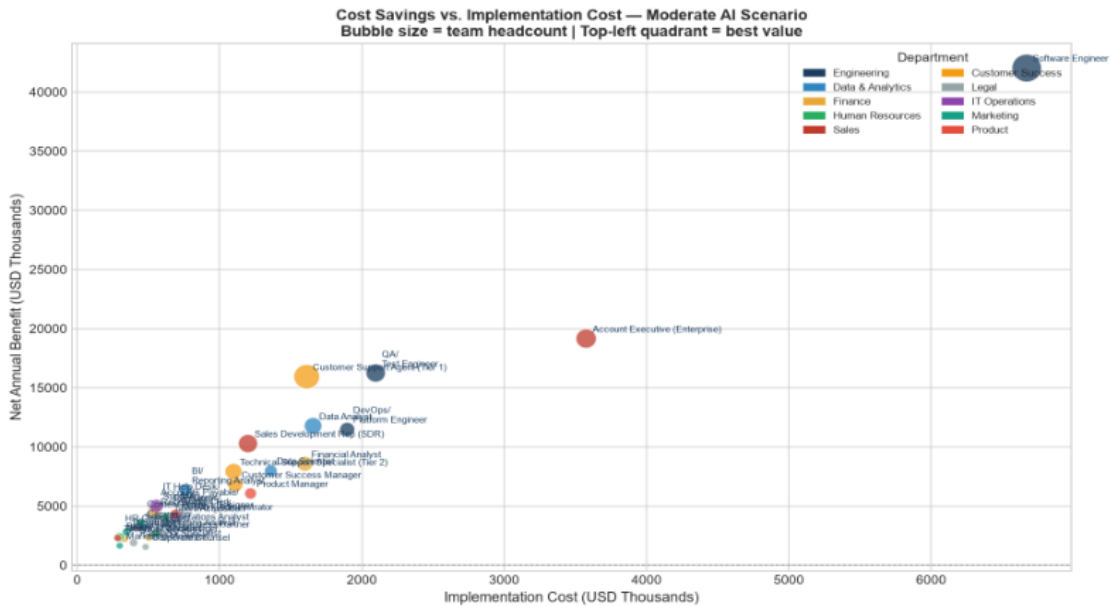
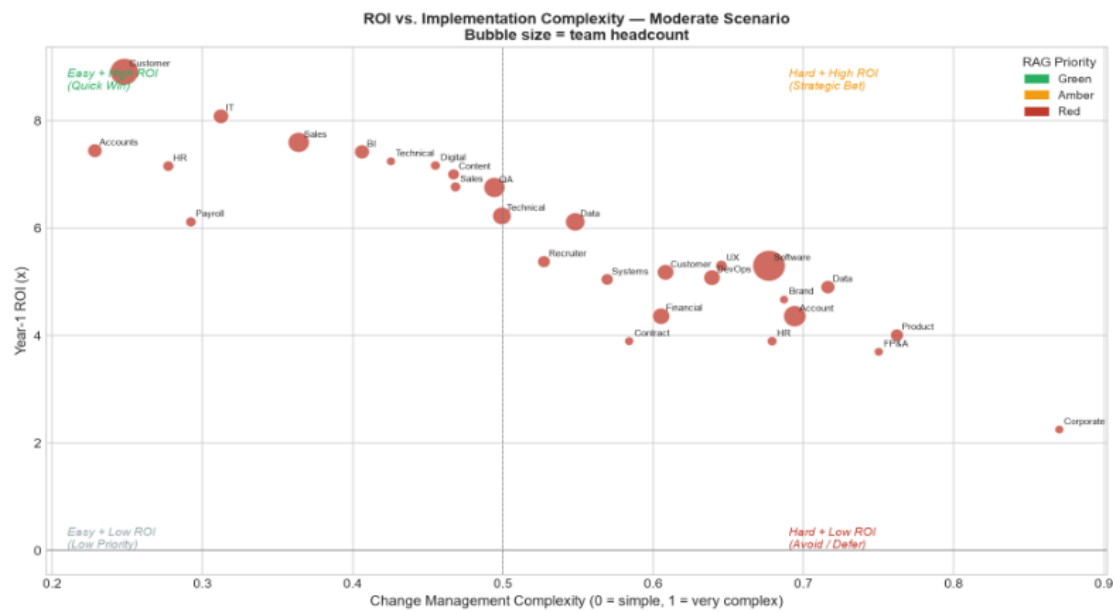


Fig 7: Savings vs Implementation Cost Bubble Chart (Bubble size = Headcount) — Identify Quick Wins in upper-left

07 | Return on Investment & Financial Justification



Top 10 Roles by Year-1 ROI (Moderate Scenario)

Role	Department	Impl. Cost	Net Benefit	Year-1 ROI	Payback (mo)
Customer Support Agent (Tier 1)	Customer Success	1609K	15958K	8.9x	1.2
IT Help Desk / Support	IT Operations	556K	5054K	8.1x	1.3
Sales Development Rep (SDR)	Sales	1200K	10315K	7.6x	1.4
Accounts Payable / Receivable Clerk	Finance	542K	4581K	7.5x	1.4
BI / Reporting Analyst	Data & Analytics	759K	6391K	7.4x	1.4
Technical Writer	Product	281K	2317K	7.2x	1.5
Digital Marketing Analyst	Marketing	347K	2835K	7.2x	1.5
HR Operations / Admin	Human Resources	290K	2366K	7.2x	1.5
Content Writer / Copywriter	Marketing	445K	3562K	7.0x	1.5
Sales Operations Analyst	Sales	427K	3318K	6.8x	1.5

08 | Risk Assessment & Strategic Decision Matrix

High savings mean nothing if AI deployment creates legal or operational failures. We compute a composite risk score:
Risk = 0.40×Quality Risk + 0.30×Regulatory Sensitivity + 0.20×Customer Impact + 0.10×Change Mgmt Complexity.

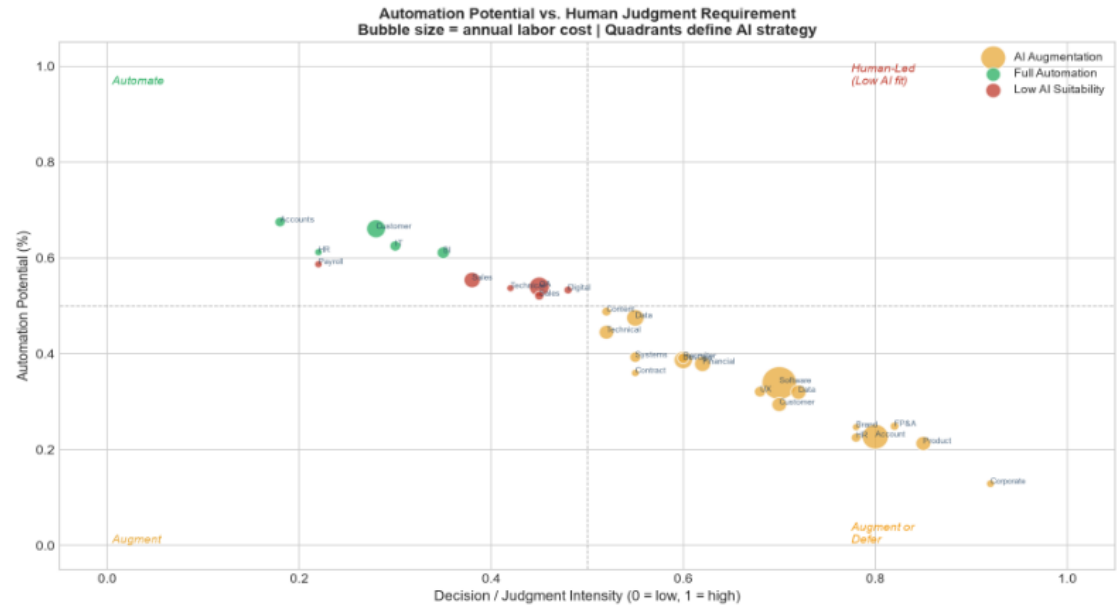


Fig 9: Automation Potential vs Human Judgment Requirement — Roles in top-left quadrant are ideal Full Automation candidates

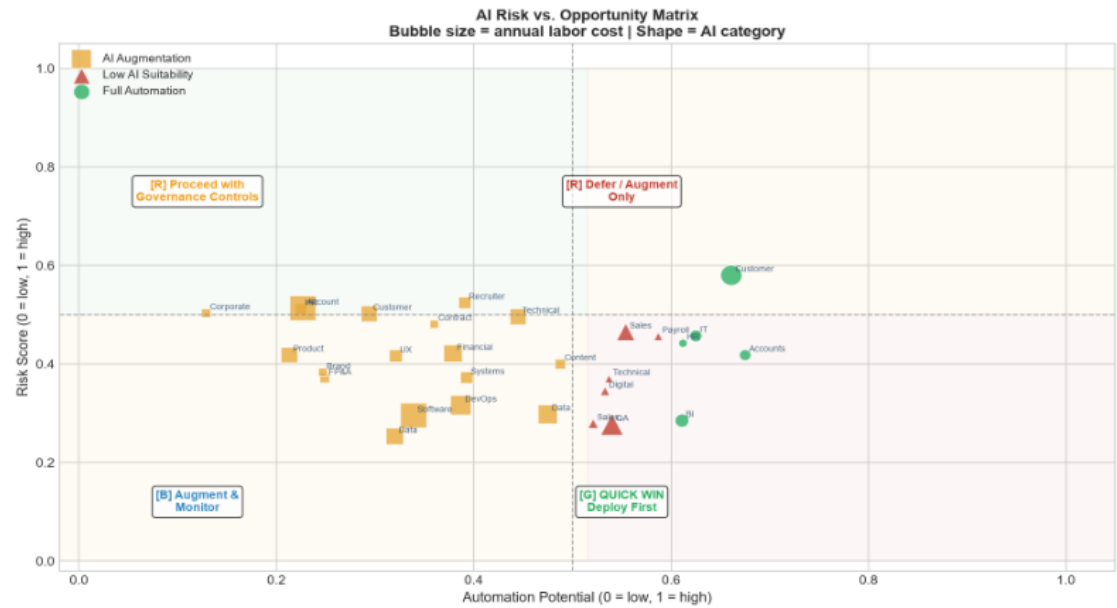


Fig 10: 2×2 Risk vs Opportunity Matrix — Quadrant classification drives deployment strategy

Quadrant Strategy Guide

08 | Risk Assessment & Strategic Decision Matrix

Quadrant Strategy Guide

Quadrant	Criteria	Strategy	Example Roles
Quick Win (Green)	Auto Potential $\geq 50\%$; Risk Score $< 40\%$	Deploy immediately in Phase 1	AP/AR Clerk; IT Help Desk; QA Engineer
Careful Deploy (Amber)	Auto Potential $\geq 50\%$; Risk Score $\geq 40\%$	Deploy with governance guardrails and human oversight	Customer Support T1; SDR
Augment First (Blue)	Auto Potential $< 50\%$; Risk Score $< 40\%$	Deploy Copilot-style tools; no FTE reduction	Software Engineer; Data Analyst; Tech Writer
Defer / Manual (Red)	Auto Potential $< 50\%$; Risk Score $\geq 40\%$	AI as background research only; human decision ownership	Corporate Counsel; HR BP; Product Manager

09 | EDA & Correlation Analysis

Before building the scoring model, we ran a comprehensive EDA to understand the underlying relationships in the workforce data. The correlation heatmap below identifies which role characteristics are statistically associated with higher automation potential.

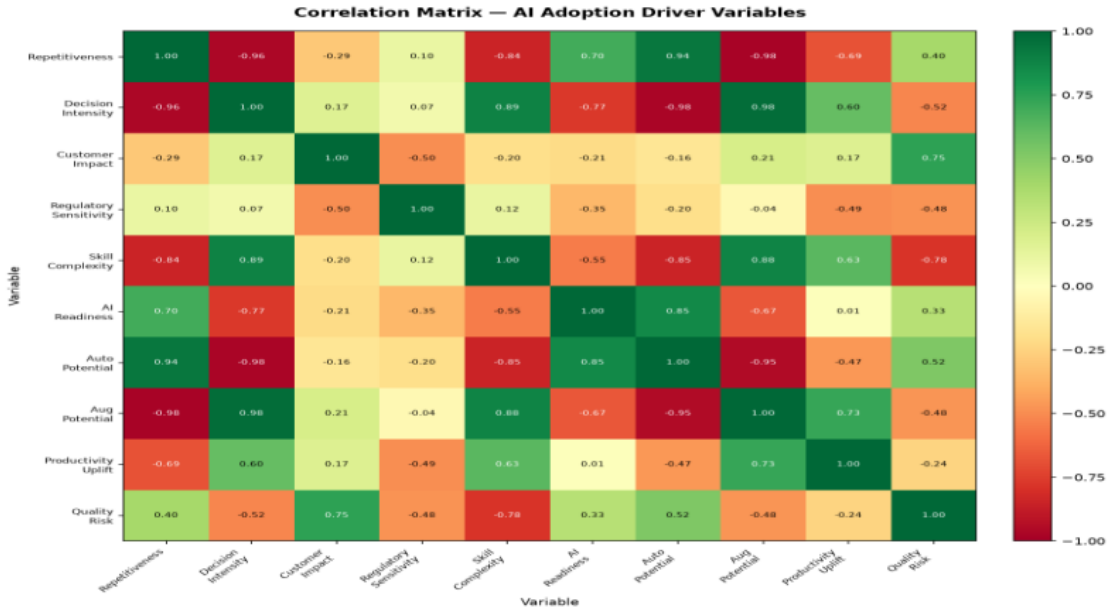


Fig 11: Correlation Heatmap — Pearson correlation between all quantitative features and automation potential

Key Statistical Correlations with Automation Potential

Factor	Pearson r	Direction	Interpretation
Task Repetitiveness	+0.937	Positive	Strongest driver — repetitive tasks are most automatable
AI Maturity / Readiness	+0.854	Positive	Mature AI tooling enables higher automation confidence
Current Error Rate	+0.773	Positive	Error-prone tasks (data entry) benefit most from automation
Decision Intensity	-0.975	Negative	Strongest barrier — judgment-heavy roles cannot be automated
Skill Complexity	-0.849	Negative	Specialised expertise reduces automation potential
Avg. Annual Salary	-0.846	Negative	High pay correlates with low automatable tasks (expertise premium)

10 | Phased Implementation Roadmap

Based on AAPS ranking, risk classification, and financial payback analysis, we recommend a three-phase, 24-month deployment roadmap. Phase assignment is driven by the composite priority index: **priority = automation_potential × (1 – risk_score) × addressable_cost_pool**.

Phase	Timeline	Focus	Roles	Expected Savings
Phase 1	0–8 mo	Quick Wins & Full Auto	Customer Support T1, AP/AR Clerk, IT Help Desk, QA/Test Engineer, BI/Reporting Analyst	\$18–22M/yr
Phase 2	8–16 mo	Careful Deploy	Tech Writer, Content Writer, SDR, Sales Ops Analyst, HR Ops Admin, Payroll Specialist	\$15–18M/yr
Phase 3	16–24 mo	Complex Augmentation	Software Engineer, Data Analyst, DevOps, Data Scientist, Customer Success Manager	\$22–26M/yr
Defer	Post-24 mo	Human-Led AI Assist	Corporate Counsel, HR BP, Product Manager, Account Exec (Enterprise), FP&A Manager, Brand Mgr	\$3–5M/yr (augment only)

Governance & Risk Guardrails

- All Phase 1 Full Automation deployments require a 3-month parallel-run period before live cutover.
- Customer-facing roles (Customer Support, Sales) require human escalation path to remain active.
- Regulatory roles (Legal, Finance) must maintain human sign-off on all AI-generated outputs.
- Quality metrics (error rate, CSAT, SLA) to be tracked monthly for first 12 months post-deployment.
- A Change Management programme covering all impacted FTE must be launched 2 months before Phase 1.

11 | AI Cost Viability — Is AI Actually Cheaper Than Humans?

A critical question many organisations overlook: **AI tooling, API costs, and implementation expenses can exceed the labour savings — especially for high-judgment, low-volume roles.** This section explicitly quantifies when AI is more expensive than the human workforce it replaces, and how sensitive that conclusion is to rising AI costs (e.g., LLM API price changes, vendor license inflation).

Methodology

For each role under the Moderate scenario (65% adoption), we compare:

- **Human Cost Replaced** = total_labor_cost × automation_potential × 65%
- **Total AI Annual Cost** = ongoing_tooling_cost + (implementation_cost ÷ 3 years amortised)
- **Cost Efficiency Ratio** = Human Cost Replaced ÷ Total AI Annual Cost
- Ratio > 1.0: AI saves more than it costs ✓ | Ratio < 1.0: AI costs more than it saves ✗
- **Cost-Parity Adoption Rate** = minimum adoption % at which AI just breaks even on cost



Cost Efficiency Ratio by Role (Human Savings ÷ AI Cost)

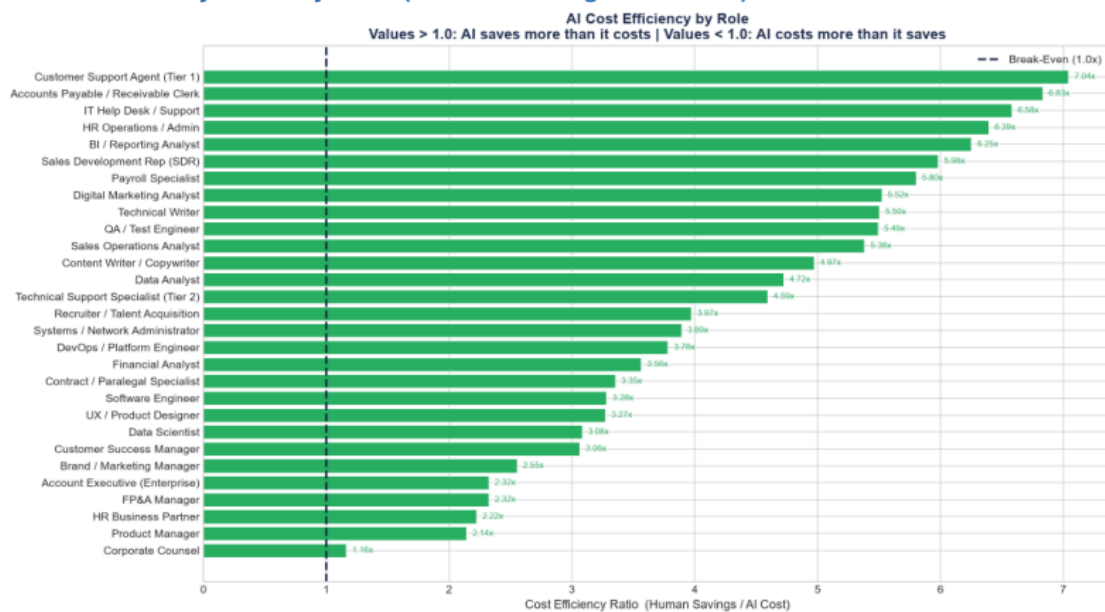


Fig 12: Cost Efficiency Ratio — Green bars (>1.0x) mean AI saves more than it costs. All 29 roles are viable at Moderate adoption, but Corporate Counsel is barely above break-even at 1.16x.

Minimum Adoption Rate Needed For AI to Break Even

This is the key risk metric: if adoption falls below this threshold (due to employee resistance, poor implementation, or tool underuse), the AI deployment loses money.

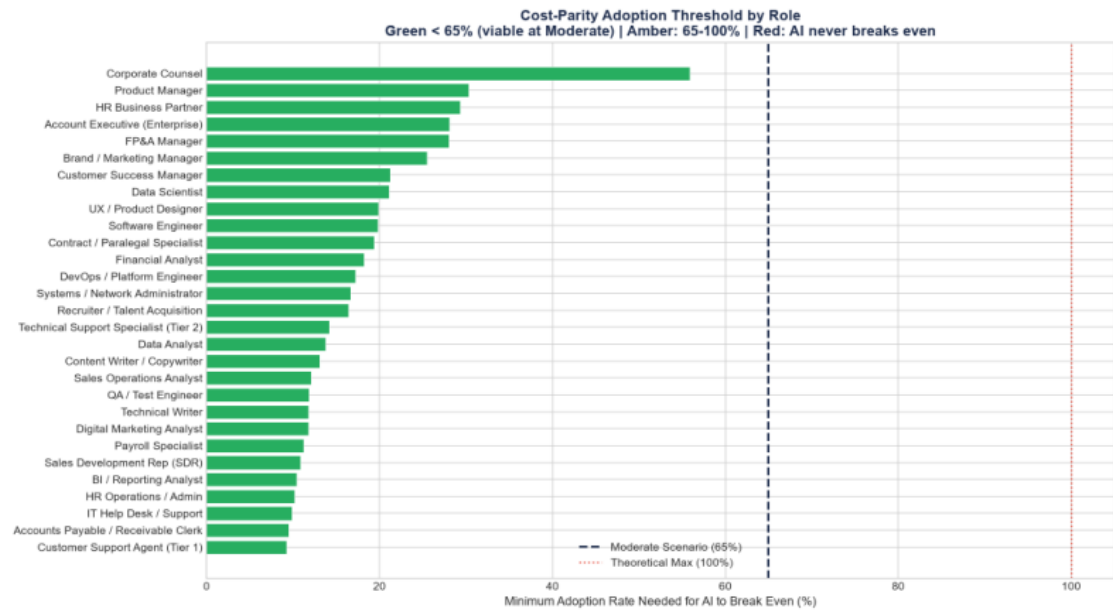


Fig 13: Cost-Parity Adoption Threshold — Green roles break even well below 65% (safe). Corporate Counsel requires 55.9% adoption to just cover costs — and has an automation potential of only 13%.

11 | AI Cost Viability (continued) — Sensitivity to Rising AI Costs

A real risk faced by enterprises today: **LLM API costs, SaaS license fees, and maintenance expenses are not fixed.** OpenAI, Anthropic, and other vendors have changed pricing multiple times. This sensitivity analysis shows how many roles flip from 'AI saves money' to 'AI costs more' as ongoing costs rise:

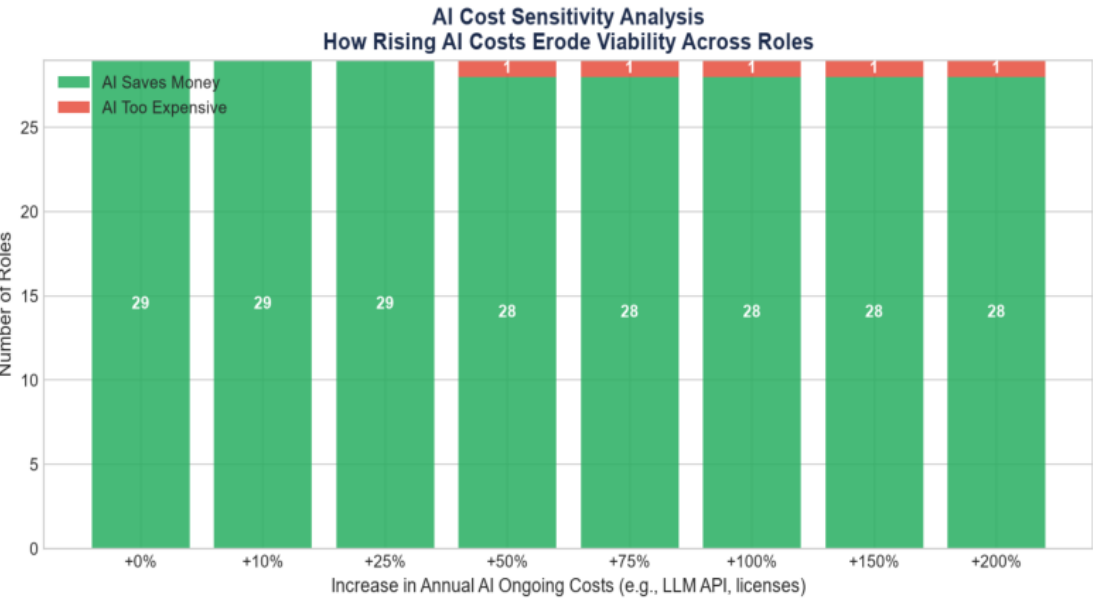


Fig 14: AI Cost Sensitivity — The portfolio is highly resilient. Even a +200% increase in ongoing AI costs only makes 1 role unviable (Corporate Counsel). All other 28 roles remain cost-effective.

Key Insight: The portfolio is robust to AI cost inflation. Because most roles have Cost Efficiency Ratios of 3x–7x, ongoing costs would need to rise by 200–600% before the majority of deployments become uneconomical. The exception is **Corporate Counsel (1.16x)** — a 17% rise in AI costs makes it unviable, which reinforces the recommendation to defer automation for this role.

Human vs AI Cost Per Productive Hour

The most direct comparison: what does it cost to get one hour of productive output from a human vs. from AI? This fully-loaded calculation includes amortised implementation costs and ongoing tooling fees.

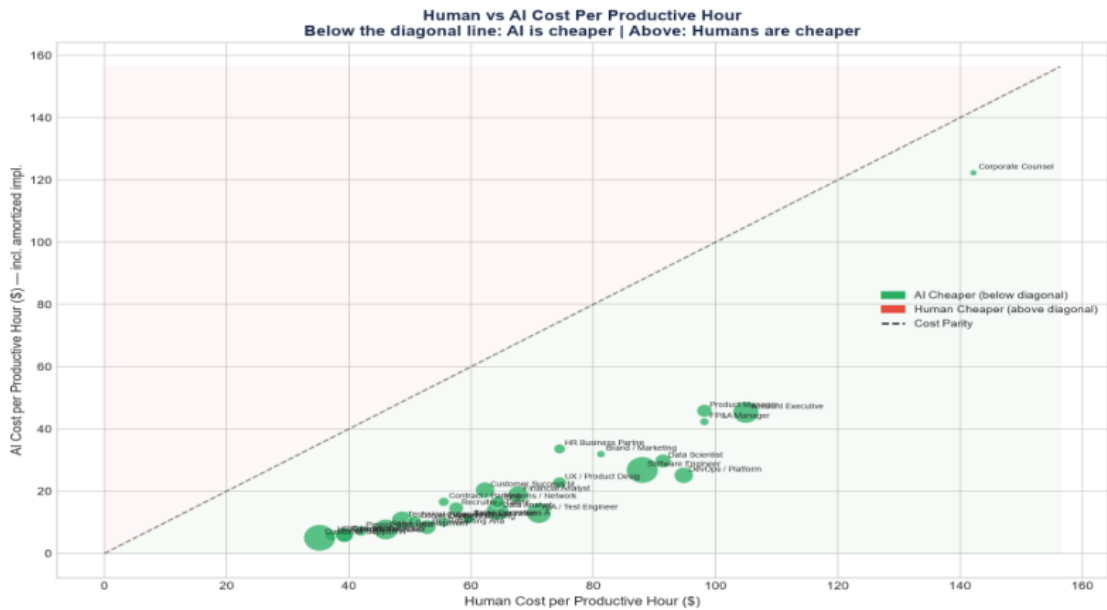


Fig 15: Human vs AI Cost Per Hour — Points below the diagonal = AI is cheaper per hour. All roles except Corporate Counsel show AI delivering the same output at a fraction of the human hourly cost.

AI Cost Viability — Role Summary Table

Role	Dept	Efficiency Ratio	Break-Even Adoption	AI Cheaper?	Human \$/Hr	AI \$/Hr
Customer Support Agent (Tier 1)	Customer Success	7.04	9.0%	YES	\$35.21	\$5.0
Accounts Payable / Receivable Clerk	Finance	6.83	10.0%	YES	\$39.27	\$5.75
IT Help Desk / Support	IT Operations	6.58	10.0%	YES	\$39.27	\$5.97
HR Operations / Admin	Human Resources	6.39	10.0%	YES	\$37.24	\$5.83
BI / Reporting Analyst	Data & Analytics	6.25	10.0%	YES	\$52.81	\$8.45
Sales Development Rep (SDR)	Sales	5.98	11.0%	YES	\$46.04	\$7.7
Payroll Specialist	Finance	5.8	11.0%	YES	\$41.98	\$7.24
Digital Marketing Analyst	Marketing	5.52	12.0%	YES	\$55.52	\$10.06
Technical Writer	Product	5.5	12.0%	YES	\$59.58	\$10.83
QA / Test Engineer	Engineering	5.49	12.0%	YES	\$71.09	\$12.96
Sales Operations Analyst	Sales	5.38	12.0%	YES	\$59.58	\$11.08
Content Writer / Copywriter	Marketing	4.97	13.0%	YES	\$50.78	\$10.22
Data Analyst	Data & Analytics	4.72	14.0%	YES	\$64.32	\$13.63
Technical Support Specialist (Tier 2)	Customer Success	4.59	14.0%	YES	\$48.75	\$10.63
Recruiter / Talent Acquisition	Human Resources	3.97	16.0%	YES	\$57.55	\$14.51
Systems / Network Administrator	IT Operations	3.89	17.0%	YES	\$64.32	\$16.51
DevOps / Platform Engineer	Engineering	3.78	17.0%	YES	\$94.79	\$25.09
Financial Analyst	Finance	3.56	18.0%	YES	\$67.71	\$18.99
Contract / Paralegal Specialist	Legal	3.35	19.0%	YES	\$55.52	\$16.56

Role	Dept	Efficiency Ratio	Break-Even Adoption	AI Cheaper?	Human \$/Hr	AI \$/Hr
Software Engineer	Engineering	3.28	20.0%	YES	\$88.02	\$26.81
UX / Product Designer	Product	3.27	20.0%	YES	\$74.48	\$22.75
Data Scientist	Data & Analytics	3.08	21.0%	YES	\$91.41	\$29.7
Customer Success Manager	Customer Success	3.06	21.0%	YES	\$62.29	\$20.35
Brand / Marketing Manager	Marketing	2.55	26.0%	YES	\$81.25	\$31.9
FP&A Manager	Finance	2.32	28.0%	YES	\$98.18	\$42.31
Account Executive (Enterprise)	Sales	2.32	28.0%	YES	\$104.95	\$45.32
HR Business Partner	Human Resources	2.22	29.0%	YES	\$74.48	\$33.59
Product Manager	Product	2.14	30.0%	YES	\$98.18	\$45.82
Corporate Counsel	Legal	1.16	56.0%	YES	\$142.19	\$122.3

12 | Conclusion & Recommendation

The data conclusively supports a structured, phased rollout of AI automation and augmentation tooling across the enterprise. The analysis demonstrates that — with the right sequencing — the organisation can capture significant value while managing operational, quality, and regulatory risk.

\$58.5M Recommended Net Annual Savings	7.5 mo Capital Payback Period (Moderate)	\$108.7M 3-Year NPV @ 10% Hurdle Rate	Moderate Recommended Scenario
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Final Recommendations

- **Prioritise Phase 1 immediately:** Customer Support T1, AP/AR Clerks, IT Help Desk, QA Engineers, and BI Analysts offer the fastest payback with the lowest risk. These roles have mature AI tooling available today.
- **Adopt the Moderate Scenario (65% capture):** Provides the best risk-adjusted return. The Aggressive scenario requires organisational change maturity that takes time to build; over-deployment early increases failure risk.
- **Invest in Developer Augmentation in Phase 3:** Software Engineers are the highest-dollar AAPS priority. Deploying Copilot tools to 420 engineers at full scale can unlock \$22–26M in net output value annually.
- **Do NOT automate Corporate Counsel or HR Business Partners:** AAPS = 0.065 and 0.211 respectively. The regulatory and reputational risk of automating these roles is not justified at any adoption rate.
- **Build a Change Management function:** Sensitivity analysis identifies that mid-tier roles (SDRs, Payroll Specialists, Data Scientists) are sensitive to weight assumptions, meaning leadership buy-in to the scoring methodology is critical before Phase 2 deployment decisions.

Methodology Note: All data is synthetic and derived from publicly available benchmarks (BLS OES 2023, Glassdoor, McKinsey Global Institute, MIT/Stanford GitHub Copilot studies, Oxford Martin School). Headcounts, exact salaries, and assumption parameters are illustrative. This report does not represent any real company's data or strategic plans.